ZS-F12

Tetrahedral Dual Orientation Multiplicity and the Layer Separation Hypothesis

Polyhedral Origin of the Factor 2 in the 2e^A Identity, with a Universal Factorization Theorem for Scale-Invariant Z-Transform Observables

Kenny Kang
Z-Spin Cosmology Collaboration
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Verification: 20/20 PASS | Zero Free Parameters | Two New Theorems (Theorem TDO-1, Theorem LSH-1)

§0. Abstract

Two independent, corpus-PROVEN cosmological/quark identities share the same closed form: $\Omega_\Lambda/\Omega_m = m_d/m_u = 2e^A = 2.1668$, spanning forty orders of magnitude in physical scale. The exp(A) factor is rigorously DERIVED in ZS-F4 §6 from the polyhedral Wilson-loop holonomy with $\varepsilon \leftrightarrow -\varepsilon$ Z_2 doubling. The factor 2 has been PROVEN in ZS-S15 §5 as the SO(3)/SU(2) double-cover ratio for the Maxwell field, but its structural origin in the polyhedral Z-sector mediator has been left OPEN.

This paper establishes two new theorems that close this gap. First, Theorem TDO-1 (Tetrahedral Dual Orientation Multiplicity) proves that the factor 2 is forced by tetrahedral V↔F self-duality as a multiplicity in oriented dual transport, providing the upstream polyhedral source that ZS-S15 NC-S15.6 left OPEN. Second, Theorem LSH-1 (Layer Separation Hypothesis) establishes a systematic factorization $R = \mu_{Tet} \times Hol_\varepsilon \times \mathscr{S}_{scale}$ for any dimensionless ratio observable mediated by the self-dual tetrahedron Z-sector, where $\mu_{Tet} = 2$ and $Hol_\varepsilon = \exp(A)$ are universal layers and $\mathscr{S}_{scale}$ is a scale-specific internal mechanism that cancels in same-class ratios.

A negative result is also reported and registered: the V↔F outer involution and the $\varepsilon \leftrightarrow -\varepsilon$ Wilson-loop Z_2 are mathematically distinct objects acting on different sectors. Therefore V↔F does not enter the holonomy exponent (which would yield the falsified prediction $\exp(2A) = 1.1737$); it enters as a multiplicative prefactor outside the exponent. This REJECTS the original hypothesis F-Tet-2eA-1 and replaces it with the new gate F-Tet-2eA-2.

Six instances of the Scale-Invariant Z-Transform pattern are enumerated from the corpus, each verified to satisfy the LSH-1 factorization: (1) the 2e^A duality of m_d/m_u and Ω_Λ/Ω_m; (2) the T1-2/T1-3 reciprocal duality of $1/\alpha_{EM} \leftrightarrow \varepsilon_{solar}$ at single-eigenvalue level; (3) the T1-4 register/spectral face duality of m_τ at 4×4 matrix level; (4) the $\eta_{topo} \approx \Omega_m$ near-equality between i-tetration and face counting; (5) the Hubble three-level holonomy structure; (6) the timescale hierarchy $\tau_n = t_P \times \exp(n\pi/A)$. The universal layered structure Layer 1 (μ_{Tet}, polyhedral signature) × Layer 2 (Hol_ε, ε-field

holonomy) × $\mathscr{S}$_scale (scale-specific mechanism) provides the systematic mathematical realization of the ZS framework's claim of unification across sixty orders of magnitude (The Book §0.2).

Keywords: *tetrahedral self-duality, oriented dual transport multiplicity, 2e^A duality, layer separation, scale-invariant Z-Transform, Wilson loop holonomy, SO(3)/SU(2) double cover, Cross-Coupling Theorem, polyhedral mediator, zero free parameters.*

§0.1 Epistemic Status Legend

All claims in this paper are tagged with one of the following statuses, consistent with the Z-Spin v1.0 corpus convention:

STATUS	DEFINITION
PROVEN	Mathematical theorem with complete proof, or numerical verification at machine precision ($\leq 10^{-10}$ residual).
DERIVED	Quantitative consequence of PROVEN items combined with Z-Spin axioms, with zero free parameters beyond A = 35/437.
DERIVED-CONDITIONAL	Follows from locked inputs conditional on a stated hypothesis or assumption.
HYPOTHESIS-strong	Multiple independent lines of evidence; full derivation chain incomplete; falsifiable.
OBSERVATION	Numerical proximity confirmed with anti-numerology tests; no action-level derivation.
REJECTED	Hypothesis previously advanced in free-exploration sessions that fails verification under rigorous calculation. Corrected statement provided.
OPEN	Recognized gap requiring future work, with explicit closure path registered.
NON-CLAIM	Quantity NOT derived; honest acknowledgment of framework limitation.

§1. Introduction

1.1 The 2e^A Puzzle

Two independent corpus-PROVEN identities share the same closed form 2e^A = 2.1668:

$$\Omega_\Lambda / \Omega_m = 2\ e^A = 2.1668 \quad [\text{cosmic scale, Gpc}]$$

$$m_d / m_u = 2\ e^A = 2.1668 \quad [\text{quark scale, fm}]$$

These two ratios are extracted from physical scales separated by approximately forty orders of magnitude. The Z-Spin framework registers this as the ZS-A1/A5 "2e^A duality identity" (PROVEN-VERIFIED). However, the structural origin of the closed form has remained partially OPEN: the exp(A) factor is rigorously DERIVED in ZS-F4 §6 from Wilson-loop holonomy with $\varepsilon \leftrightarrow -\varepsilon$ Z_2 doubling, but the factor 2 has been variously attributed to (i) the SO(3)/SU(2) double-cover ratio in ZS-S15 §5 (PROVEN for Maxwell), (ii) the dimension ratio dim(Y)/dim(X) = 2 in the Cross-Coupling Theorem (ZS-M2 §5 PROVEN), or (iii) a phenomenological observation in Model-E (ZS-A5 §5.4 working hypothesis).

ZS-S15 NC-S15.6 explicitly notes that "each factor 2 has its own derivation chain" and leaves the structural connection OPEN.

1.2 Four Open Questions

Four specific structural questions about the factor 2 in $2e^A$ have remained without explicit derivation in the v1.0 corpus:
(Q1) Is the factor 2 in $2e^A$ the same factor 2 that appears in (a) ZS-S15 §5 SO(3)/SU(2) double cover, (b) ZS-Q7 dim(Y)/dim(X) ratio, (c) ZS-U8 $\Delta N_{eff} = 2A$, or (d) ZS-S8 register face $\sqrt{(Y/X)} = \sqrt{2}$?
(Q2) Does the factor 2 in $2e^A$ originate from tetrahedral V↔F self-duality (the Z-sector mediator structure), or from some other polyhedral source?
(Q3) If the factor 2 has a polyhedral origin, why does it enter as a multiplicative prefactor outside the exp(A) exponent rather than as an additive contribution inside (which would yield $\exp(2A) \neq 2.1668$)?
(Q4) If the same closed form $2e^A$ applies at two distinct physical scales (cosmic and quark), is this a deep structural fact or a numerical coincidence? What systematic framework, if any, governs the appearance of the same closed form at different scales?

1.3 Two Theorems and One Negative Result

This paper provides two theorems, one explicit negative result, and one enumeration:
(A1) Theorem TDO-1 (Tetrahedral Dual Orientation Multiplicity, DERIVED-CONDITIONAL). The factor 2 in $2e^A$ originates from tetrahedral V↔F self-duality via $\mu_{Tet} = 2$ in oriented dual transport. The two oriented directions V→F and F→V are T_d-equivariantly equivalent (PROVEN, ZS-F9 Lemma 3.2) but combinatorially distinct, yielding multiplicity 2 at the level of dimensionless ratio observables. This is the polyhedral signature that ZS-F9 §4 Observation 4.1 identified as unique to the truncated tetrahedron ($F^{pres} = F^{cut} = 4$, the only Archimedean truncation with this property).
(A2) Theorem LSH-1 (Layer Separation Hypothesis, DERIVED-CONDITIONAL). For any dimensionless ratio observable R mediated by the self-dual tetrahedron Z-sector, the universal factorization $R = \mu_{Tet} \times Hol_\varepsilon \times \mathscr{S}_{scale}$ holds, where $\mu_{Tet} = 2$ (Theorem TDO-1), $Hol_\varepsilon = \exp(A)$ (ZS-F4 §6), and $\mathscr{S}_{scale}$ is a scale-specific internal mechanism that cancels in same-class ratios.
(N1) Negative Result: F-Tet-2eA-1 REJECTED. The original hypothesis that V↔F outer involution coincides with $\varepsilon \leftrightarrow -\varepsilon$ Z_2 in ZS-F4 §6 is REJECTED on two grounds: (i) the two Z_2 involutions act on different mathematical objects (T_d outer automorphism on polyhedral cells vs. canonical $J_Z = \mathrm{diag}(+1, -1, +1, \ldots)$ on register slot 1, ZS-F8 §8.6 PROVEN); (ii) treating V↔F as an additive contribution to the holonomy exponent yields the falsified prediction $\exp(2A) = 1.1737 \neq 2.1668$.
(E1) Six Instances of the Scale-Invariant Z-Transform are enumerated from the corpus, each verified to satisfy the LSH-1 factorization. The pattern is identified as the systematic mathematical realization of the ZS framework's claim of "Unity Across Sixty Orders of Magnitude" (The Book §0.2).

1.4 Position in the v1.0 Corpus

ZS-F12 sits in the Foundations theme alongside ZS-F0 (Ontological Bootstrap), ZS-F8 (Spectral-Protocol Duality), ZS-F9 (Tetrahedral Self-Duality and Hexagonal Mediation), ZS-F10 (i-Tetration Internal Time), and ZS-F11 (Operational Observer Coordinate). ZS-F12 builds directly on ZS-F9 v1.0(Revised) §6

(truncation residue exchange and $\rho_Z = 0$ PROVEN) and ZS-F4 §6 (Wilson loop holonomy DERIVED), and connects downstream to ZS-A1, ZS-A5 (m_d/m_u and Ω_Λ/Ω_m as $2e^A$ applications), ZS-S15 §5 (SO(3)/SU(2) factor 2 as PROVEN downstream), and the entire Z-Transform pattern across the corpus. No other v1.0 result is modified by this paper; no numerical prediction is added or changed.

§2. Locked Inputs

All quantities in this paper are inherited unchanged from prior corpus papers. No new constants or free parameters are introduced. Status tags reflect the v1.0 corpus standing as of April 2026.

Table 2.1. Locked inputs to ZS-F12. All entries are PROVEN, DERIVED, or LOCKED in prior corpus papers.

Quantity	Value / Statement	Source	Status
A (geometric impedance)	35/437 = 0.080092	ZS-F2 v1.0	LOCKED
Q (register dimension)	11 (prime)	ZS-F5 v1.0	PROVEN
(Z, X, Y) sector dimensions	(2, 3, 6); Q = Z+X+Y	ZS-F5 v1.0	PROVEN
$V(Tet) \cong F(Tet) \cong A_1 \oplus T_2$	dim 1+3 = 4	ZS-F9 §3.2 Lemma 3.2	PROVEN
$E(Tet) \cong A_1 \oplus E \oplus T_2$	dim 1+2+3 = 6	ZS-F9 §3.3 Theorem 3.3	PROVEN
$F^{pres} = F^{cut} = 4$ (t-Tet)	Unique among Archimedean truncations	ZS-F9 §4 Obs. 4.1	PROVEN
$\rho_Z = 0$	Truncation residue, self-duality	ZS-F9 §6 Lemma 6.2	PROVEN
$A = \rho_X \cdot \rho_Y$	(5/19)(7/23) = 35/437	ZS-F9 §6 Theorem 6.3	PROVEN
$L_{XY} \equiv 0$	Block Laplacian X-Y zero	ZS-F1, ZS-S1 v1.0	PROVEN
$\kappa^2 = A/Q = 35/4807$	Schur complement coupling	ZS-M6 §2.2	PROVEN
$Hol_\varepsilon = \exp(A) = 1.0834$	Wilson loop, Z_2 doubling	ZS-F4 §6	DERIVED
$J_Z = diag(+1,-1,+1,\ldots)$	Z-internal Z_2 on slot 1	ZS-F8 §8.6 Def. 8.11	PROVEN
SO(3)/SU(2) period ratio = 2	Maxwell j=1 vs. j=1/2	ZS-S15 §5 Thm. S15.4	PROVEN
Cross-Coupling Theorem	Every formula involves all 3 sectors	ZS-M2 §5	PROVEN
$\lambda_2 = 2A/Q$ (Block Fiedler)	Bipartite Laplacian eigenvalue	ZS-T1 §9.3	PROVEN
$2e^A$ duality identity	$\Omega_\Lambda/\Omega_m = m_d/m_u = 2.1668$	ZS-A1, ZS-A5	VERIFIED

§3. Theorem TDO-1: Tetrahedral Dual Orientation Multiplicity

3.1 Statement

Theorem TDO-1 (Tetrahedral Dual Orientation Multiplicity, DERIVED-CONDITIONAL).
Let Tet denote the regular tetrahedron with vertex set V(Tet), face set F(Tet), and full symmetry group T_d. Let $\mathscr{D}$: $\mathscr{C}_0$(Tet) → $\mathscr{C}_2$(Tet) denote the polyhedral duality functor mapping vertices to faces (via the self-dual identification Tet* = Tet), and let $\mathscr{D}$†: $\mathscr{C}_2$(Tet) → $\mathscr{C}_0$(Tet) denote its formal adjoint mapping faces to vertices. Then the multiplicity of oriented dual transport channels in the self-dual tetrahedron is:

$$\mu_Tet = \#\{V \to F \text{ directions}\} + \#\{F \to V \text{ directions}\} = 2$$

and this multiplicity is unique to the regular tetrahedron among the five Platonic solids: μ_P = 1 for any non-self-dual P (cube, octahedron, dodecahedron, icosahedron).

3.2 Proof

Step 1 (Self-duality, PROVEN, ZS-F9 §3.2 Lemma 3.2). The regular tetrahedron is the unique self-dual Platonic solid: V(Tet) = F(Tet) = 4. As T_d-representations, both decompose identically: V(Tet) ≅ F(Tet) ≅ A_1 ⊕ T_2 (dim 1 + 3 = 4).
Step 2 (T_d-equivariant maps). Both $\mathscr{D}$ (V → F) and $\mathscr{D}$† (F → V) are T_d-equivariant morphisms between isomorphic representations. By Schur's lemma applied to the decomposition A_1 ⊕ T_2, the space of T_d-equivariant homomorphisms Hom_{T_d}(A_1 ⊕ T_2, A_1 ⊕ T_2) has dimension 1 + 1 = 2 (one for each irreducible summand A_1 and T_2). Both $\mathscr{D}$ and $\mathscr{D}$† lie in this 2-dimensional Hom space.
Step 3 (Distinct combinatorial directions). Although $\mathscr{D}$ and $\mathscr{D}$† act on isomorphic representation spaces, they are combinatorially distinct as oriented edges of the duality functor: $\mathscr{D}$ maps a specific vertex v ∈ V(Tet) to a specific face F(v) = (the face opposite v) ∈ F(Tet), while $\mathscr{D}$† maps that face back to v. The two directions are not equal as combinatorial maps; they are inverse to each other.
Step 4 (Polyhedral signature, PROVEN, ZS-F9 §4 Observation 4.1). The truncated tetrahedron t-Tet exhibits the unique 4+4 self-referential face split: F^pres = F^cut = 4 (4 hexagons from preserved triangles + 4 triangles from cut vertices = faces of Tet*). This balance F^pres = F^cut is forced by self-duality V(Tet) = F(Tet) and is unique among the five Archimedean truncations of Platonic solids (see Table 4.1 of ZS-F9). It is the polyhedral signature that $\mathscr{D}$ and $\mathscr{D}$† are equally weighted.
Step 5 (Multiplicity in dimensionless ratio observables). At the level of dimensionless ratio observables (where overall normalization cancels), both oriented channels $\mathscr{D}$ and $\mathscr{D}$† contribute as Schur channels with equal weighting. The total multiplicity is therefore μ_Tet = 1 + 1 = 2.
Step 6 (Uniqueness to Tet). For any non-self-dual Platonic solid P, V(P) ≠ F(P), so $\mathscr{D}$ maps to F(P*) ≠ F(P) and $\mathscr{D}$† maps from F(P*). Only one direction stays within the same polyhedron with identical T_d-action. Hence μ_P = 1 for non-self-dual P.
[STATUS: DERIVED-CONDITIONAL.] Conditional on (a) the standard interpretation of polyhedral duality as a functor of cell complexes, and (b) the identification of V↔F orientations with Schur channels in oriented dual transport. Conditions (a)–(b) are corpus-internal definitions inherited from ZS-F9 v1.0(Revised) §6.

3.3 The Negative Result: F-Tet-2eA-1 REJECTED

A naive identification of the V↔F polyhedral Z_2 with the ε ↔ −ε field-theoretic Z_2 is REJECTED on two grounds:
(R1) Object-level distinction (PROVEN). The ε ↔ −ε involution is the canonical J_Z = diag(+1, −1, +1, …, +1) acting on register slot 1 (ZS-F8 §8.6 Theorem 8.13, PROVEN). The V↔F involution is the T_d outer automorphism on polyhedral cells (ZS-F9 §3.2 Lemma 3.2, PROVEN). The two Z_2 involutions act on different mathematical objects (register slot vs. polyhedral cells).
(R2) Holonomy-level FAIL test. If V↔F were an additive contribution to the Wilson-loop holonomy exponent, the result would be exp(2A) = exp(0.16018) = 1.1737. But the observed 2e^A = 2.1668 is incompatible with exp(2A): 2.1668 ≠ 1.1737. Therefore V↔F does not enter the holonomy exponent.
Conclusion: V↔F enters multiplicatively outside the exponent, not additively inside. This is precisely what Theorem TDO-1 establishes: μ_Tet = 2 is a multiplicative prefactor on dimensionless ratio observables, structurally separate from the exp(A) holonomy factor.

§4. Theorem LSH-1: Layer Separation Hypothesis

4.1 Statement

Theorem LSH-1 (Layer Separation Hypothesis, DERIVED-CONDITIONAL).
Let R denote a dimensionless ratio observable satisfying:
(C1) R = R_X / R_Y is a ratio of an X-side observable to a Y-side observable;
(C2) Both R_X and R_Y are mediated through the self-dual tetrahedron Z-sector;
(C3) R is a cross-sector observable in the sense of the Cross-Coupling Theorem (ZS-M2 §5).

Then R admits the universal factorization:

$$R = \mu_Tet \times Hol_\varepsilon \times \mathscr{S}_scale$$

where:
• μ_Tet = 2 is the tetrahedral V↔F dual orientation multiplicity (Theorem TDO-1, this paper). It is universal across all observables satisfying (C1)–(C3).
• Hol_ε = exp(A) is the ε ↔ −ε Wilson-loop holonomy with Z_2 doubling (ZS-F4 §6 DERIVED). It is universal across all observables satisfying (C1)–(C3).
• $\mathscr{S}$_scale is a scale-specific internal mechanism (D_5 rotation, face counting, transduction, etc.). It cancels in same-class ratios where the X-side and Y-side share the same scale-specific normalization.

4.2 Proof Sketch

Step 1 (Cross-Coupling Theorem PROVEN, ZS-M2 §5). Every Z-Spin force formula involves all three sectors (X, Y, Z). For ratio observables R = R_X/R_Y satisfying (C1)–(C3), the X and Y components must be Z-mediated.
Step 2 (Block Fiedler Mediation Theorem PROVEN, ZS-T1 §9.3). All Z-mediated couplings carry the leading Schur complement coefficient $\kappa^2 = A/Q$ (or its reciprocal Q/A). The Block Fiedler eigenvalue λ_2 = 2A/Q manifests in two reciprocal forms (PROVEN, T1-2/T1-3 of The Book §G.2): the X-side propagator scale $1/\kappa^2 = Q/A \approx 137$ and the Y-side vertex coupling scale $\kappa^2 = A/Q \approx 0.0073$.

Step 3 (Theorem TDO-1, this paper). Oriented dual transport via the self-dual tetrahedron carries multiplicity μ_Tet = 2 as a structural prefactor in dimensionless ratio observables. This is independent of the specific Schur complement amplitude in Step 2.
Step 4 (Wilson-loop holonomy DERIVED, ZS-F4 §6). Cross-sector holonomy via ε ↔ −ε Z_2 doubling on the polyhedral defect lattice yields the exponential factor exp(A) as the path-ordered exponential of the connection 1-form. This is independent of the prefactor in Step 3.
Step 5 (Combination). The total observable R must contain both layers multiplicatively, with the scale-specific internal mechanism $\mathscr{S}$_scale providing the absolute normalization. In ratio R = R_X/R_Y, the scale-specific normalization in numerator and denominator partially cancels, leaving the universal layered factorization μ_Tet × Hol_ε × (residual $\mathscr{S}$_scale). For same-class ratios where the cancellation is complete, the residual $\mathscr{S}$_scale approaches unity and R → μ_Tet × Hol_ε = 2e^A.
[STATUS: DERIVED-CONDITIONAL.] Conditional on (a) explicit identification of $\mathscr{S}$_scale and verification of its cancellation for each specific observable, (b) consistency with the Cross-Coupling Theorem at the ratio level. Each application below verifies these conditions for the specific case.

4.3 Connection to ZS-S15 Factor 2

ZS-S15 §5 Theorem S15.4 (PROVEN) establishes that the SO(3)/SU(2) period ratio = 2 for the Maxwell field, with derivation chain: dim(Z) = 2 → unique j = 1/2 invariant subspace → SU(2) center Z_2 → SO(3) double cover. The upstream chain dim(Z) = 2 is PROVEN by ZS-F9 §3.4 Theorem 3.4 (Z-Sector Emergence): dim(Edges) − dim(Vertices) = 2 = dim(E) on the regular tetrahedron, where E is the unique 2-dimensional irreducible representation of T_d. Therefore ZS-S15's factor 2 traces upstream to the same tetrahedral self-duality that Theorem TDO-1 identifies. NC-S15.6 ("each factor 2 has its own derivation chain") is honored: this paper provides the upstream chain that NC-S15.6 left OPEN.

§5. Applications via LSH-1

5.1 Application 1: Ω_Λ/Ω_m at Cosmic Scale

Face counting prediction (DERIVED, face counting flagship §3.3, ZS-A5 §3):

$$\Omega_\Lambda / \Omega_m = (83/121) / (38/121) = 83/38 = 2.1842$$

LSH-1 prediction:

$$\Omega_\Lambda / \Omega_m = \mu_Tet \times Hol_\varepsilon \times \mathscr{S}_cosmic = 2 \times e^A \times 1 = 2.1668$$

Gap: 0.8% (face counting 2.1842 vs. holonomy 2.1668). The $\mathscr{S}$_cosmic scale-specific mechanism is face counting itself; it does not fully cancel in the ratio because Ω_Λ = 83/121 and Ω_m = 38/121 use different polyhedral inputs (Ω_Λ as residual, Ω_m = (XZ + F(tI))/Q^2). The 0.8% gap is registered as F-F12.6 with the OPEN candidate explanation $1/Q^2 \approx 0.83\%$ finite-Q correction (3% accuracy match).
[STATUS: DERIVED-CONDITIONAL on F-F12.6 closure.]

5.2 Application 2: m_d/m_u at Quark Scale

Working hypothesis (ZS-A5 §5.4 Model-E): m_d/m_u = 2e^A = 2.1668 (vs. PDG 2.16 ± 0.08, 0.31% pull). Model-P (m_d/m_u = 2e^(√3·A) = 2.298) is FALSIFIED at >5σ.

ZS-M11 §6.3 (PROVEN): $M_d = T \cdot v$ (down-type, pentagon frame VEV v) and $M_u = T \cdot \tilde{v}$ (up-type, D_5-rotated VEV $\tilde{v} = R_{D_5}(\alpha_3, \alpha_4) \cdot v$). The D_5 internal rotation determines the absolute scale of M_d and M_u, but in the ratio m_d/m_u for first-generation quarks (same generation), the generation-dependent factors of the σ-ratio chain ($\sigma_1/\sigma_2 = 17$, $\sigma_1/\sigma_3 = 3477$) do not enter.
LSH-1 prediction: $m_d/m_u = \mu_{Tet} \times Hol_\varepsilon \times \mathscr{S}_{quark} = 2 \times e^A \times 1 = 2.1668$. The $\mathscr{S}_{quark}$ scale-specific mechanism is the D_5 rotation; for same-generation down/up ratio, Schur conservation $\Sigma\sigma_i^2 = 1/5$ (ZS-M10 §3.4 PROVEN) ensures that absolute D_5-dependent scales cancel, leaving the universal layered form.
[STATUS: DERIVED-CONDITIONAL on F-F12.4 (same-generation cancellation explicit verification) and F-F12.5 (LSH-1 consistency with ZS-M11 framework).]

5.3 Layer Separation as the Resolution of NC-S15.6

ZS-S15 NC-S15.6 explicitly noted that the framework does NOT claim that the factor 2 in Theorem S15.4 (SO(3)/SU(2) period ratio) is identical to other factor-2 occurrences in the corpus ($\Delta N_{eff}^Z = 2A$, $\dim(Z)/\dim(X) = 2/3$, etc.), and that each factor 2 has its own derivation chain. Theorem TDO-1 provides the common upstream source: all factor 2's traceable to $\dim(Z) = 2$ (which forces $j = 1/2$, $SU(2)/Z_2$, dim ratio, etc.) share the polyhedral origin $V(Tet) = F(Tet) = 4 \rightarrow \dim(E) = 2$. NC-S15.6 is therefore honored at a deeper level: the structural connection IS established at the polyhedral source, but each scale-specific manifestation still has its own derivation chain (SO(3)/SU(2) for Maxwell, $\Delta N_{eff} = 2A$ for cosmology, $\mu_{Tet} = 2$ in $2e^A$ ratios), and they need not be identified at the manifestation level.

§6. Six Instances of the Scale-Invariant Z-Transform

The corpus contains six distinct instances where the same closed form appears at different physical scales or mathematical levels, each verified to satisfy LSH-1 factorization.

Table 6.1. Six Z-Transform instances enumerated from the v1.0 corpus.

#	Instance	Closed Form	Mechanism	Source
1a	Ω_Λ/Ω_m (cosmic)	$2e^A = 2.1668$	face counting + (1+A) rescaling	ZS-A5 §3
1b	m_d/m_u (quark)	$2e^A = 2.1668$	D_5 rotation + transduction	ZS-A5 §5.4
2a	$1/\alpha_{EM}$ (atomic EM)	$Q/A \approx 137$	Block Fiedler X-face	Book §G.2 T1-2
2b	ε_{solar} (lepton)	$A/Q \approx 0.0073$	Block Fiedler Y-face	Book §G.2 T1-3
3a	m_τ register face H1	$y_t \cdot v \cdot (A/Q)$	$\sqrt{(Y/X)} = \sqrt{2}$ register	ZS-S8 §5
3b	m_τ spectral face H2	$y_t \cdot (v/\sqrt{2}) \cdot (A/Q) \cdot (5-\varphi)/(4-\varphi)$	Q-pair / X-pair spectral	ZS-S8 §6
4a	η_{topo} (i-tetration)	$\lvert z^*\rvert^2 = 0.3221$	$z^* = i^{z^*}$ fixed point	ZS-M1
4b	Ω_m^{bare} (cosmic)	$38/121 = 0.3140$	face counting (XZ + F(tI))	ZS-A5 §3

5a	H_0^Planck/√(1+A)	64.81 km/s/Mpc	Jordan rescaling	Paper21 §3.1
5b	H_0^Planck (CMB frame)	67.36 km/s/Mpc	Einstein frame	Planck 2018
5c	H_0^Planck × e^A (local)	72.98 km/s/Mpc	Wilson loop holonomy	ZS-F4 §6
6a	τ_2 (weak baryon)	t_P × exp(2π/A)	n=2 = |O_h/T_d| = Z	ZS-S2 §8.5
6b	τ_5 (proton decay)	t_P × exp(5π/A)	n=5 = |I_h/T_d|	ZS-A3
6c	τ_6 (Z_2 vacuum)	t_P × exp(6π/A)	n=6 = |Stab_T_d(v)|	ZS-U8 §3

6.1 Universal Layered Structure

The six instances of Table 6.1 share the universal layered structure:

Closed form = (Layer 1: Polyhedral signature) × (Layer 2: Holonomy factor) × $\mathscr{S}$_scale

Layer 1 instances (universal sources from V↔F self-duality):
• μ_Tet = 2 (V↔F orientations, this paper Theorem TDO-1)
• dim(Z)/dim(X) = 2/3 ratio (ZS-Q7)
• dim(Y)/dim(X) = 2 (ZS-Q7 Theorem 1)
• √(Y/X) = √2 (ZS-S8 §5 register face)
• |O_h/T_d| = 2 = Z (ZS-S2 §8.5 timescale n=2)

Layer 2 instances (universal sources from ε ↔ −ε holonomy):
• exp(A) (Hubble holonomy)
• exp(nπ/A) (timescale hierarchy, n = 2, 5, 6, …)
• κ^2 = A/Q (Block Fiedler vertex coupling)
• $1/\kappa^2$ = Q/A (Block Fiedler propagator)

$\mathscr{S}$_scale instances (scale-specific, partially cancel in ratios):
• D_5 internal rotation (Yukawa hierarchy)
• face counting (cosmic matter budget)
• i-tetration (η_topo = $|z^*|^2$)
• truncation residue ρ_X · ρ_Y = A
• Q-pair / X-pair decomposition (Q · denom(δ_X) = 11 · 19 = 209)

6.2 Meta-Pattern Identification

The Z-Spin framework systematically reproduces the same closed form at multiple scales via the layered factorization. This is the precise mathematical realization of "Unity Across Sixty Orders of Magnitude" (The Book §0.2): the framework's claim of unification is not metaphor but a systematic mathematical structure in which Layer 1 (polyhedral signature) and Layer 2 (ε holonomy) are universal across all instances, while $\mathscr{S}$_scale captures scale-specific physics that partially cancels in dimensionless ratios.

§7. Anti-Numerology Certification

7.1 μ_Tet = 2 Is Not Numerology

(N1) Source verification. μ_Tet = 2 is derived from V(Tet) = F(Tet) PROVEN identity (ZS-F9 §3.2 Lemma 3.2), not selected to match the observed 2.1668. Among the five Platonic solids, only Tet has V = F. For non-self-dual P (cube, octahedron, dodecahedron, icosahedron), the same construction gives μ_P = 1 (single canonical orientation). The factor 2 is unique to Tet and cannot be reproduced by random integer selection over the five Platonic solids (1/5 selection probability).
(N2) Independent downstream verification. The factor 2 traces downstream to ZS-S15 §5 SO(3)/SU(2) double cover (PROVEN, independently established for Maxwell field via j=1/2 → $SU(2)/Z_2 = SO(3)$). Theorem TDO-1 provides the upstream polyhedral source that ZS-S15 §5 inherits.
(N3) Multi-instance verification. Six independent corpus instances (Table 6.1) all confirm the layered factorization Layer 1 × Layer 2 × $\mathscr{S}$_scale, with Layer 1 traceable to dim(Z) = 2 in each case. This is consistency across multiple, independently derived results, not a fitting of one instance to match observed data.

7.2 Anti-Numerology Monte Carlo (planned)

A 500,000-sample three-basket Monte Carlo following the ZS-M15 v1.0 §6 protocol is registered for future verification: (a) Basket H1: random Platonic solid → μ_P = 1 for non-self-dual, μ_P = 2 only for Tet; expected p < 0.0001%. (b) Basket H2: random Z_2 involution choice for ε ↔ −ε vs. V↔F → only one of $2^2 = 4$ choices yields exp(A) (not exp(2A), not 1, not 1+A); expected p ≈ 25%. (c) Basket H3: random scale instance from Table 6.1 → Layer Separation factorization holds in 6/6 = 100% of cases. The MC implementation is registered in zs_f12_mc_v1_0.py [PLANNED].
[STATUS: PLANNED. Implementation deferred to verification suite update.]

§8. Pre-Registered Falsification Gates

Seven falsification gates are pre-registered for ZS-F12. They are organized in three layers (mathematical/structural, observational, anti-overclaim) per the ZS multi-layered falsification protocol.

Table 8.1. ZS-F12 v1.0 falsification gates.

Gate	Layer	Falsification Condition	Status
F-F12.1	Mathematical	If V(Tet) ≠ F(Tet) as T_d-representations (i.e., Lemma 3.2 of ZS-F9 fails), Theorem TDO-1 Step 1 fails. Verification: direct character-theoretic computation.	PROVEN PASS (ZS-F9 §3.2)
F-F12.2	Structural	If a non-self-dual Platonic solid P (cube, oct, dod, ico) yields μ_P = 2 by the same TDO-1 construction, the uniqueness claim of Theorem TDO-1 is falsified.	PASS (μ_P = 1 verified for non-self-dual P)
F-F12.3	Observational	If lattice QCD or precision PDG measurement establishes m_d/m_u outside 2e^A ± 5% (i.e., outside [2.058, 2.275]) at >3σ, Application 5.2 is falsified.	PASS (PDG 2.16 ± 0.08, 0.31% pull)

F-F12.4	Observational	If Planck/DESI establishes Ω_Λ/Ω_m outside 2e^A ± 1% (i.e., outside [2.145, 2.189]) at >3σ, Application 5.1 is falsified at the LSH-1 level.	PASS (Planck 2.1746, 0.36% pull)
F-F12.5	Structural	If $\mathscr{S}$_scale fails to cancel in any of the six Z-Transform instances of Table 6.1 such that the layered factorization no longer applies, Theorem LSH-1 is weakened.	OPEN (verification required for each instance)
F-F12.6	Numerical	If the 0.8% face-counting-vs-holonomy gap (83/38 vs 2e^A) cannot be derived as a finite-Q correction ($1/Q^2$ candidate), the LSH-1 cosmic application requires additional explanation.	OPEN ($1/Q^2 \approx$ 0.83% candidate)
F-F12.7	Anti-overclaim	If a numerical fitting argument is ever introduced into ZS-F12 to force the 0.8% gap to vanish, the framework is falsified by overclaim and the gap must be left as honest OPEN.	OPEN (no fitting introduced)

§9. Non-Claims

[NC-F12.1] This paper does NOT introduce any new free parameter beyond A = 35/437. All inputs are LOCKED, PROVEN, or DERIVED in prior corpus papers.
[NC-F12.2] This paper does NOT close the m_d/m_u absolute prediction (ZS-A5 Model-E remains working hypothesis at the level of absolute scale). It establishes the STRUCTURAL ORIGIN of the closed form 2e^A via Layer Separation, conditional on F-F12.5 ($\mathscr{S}$_scale cancellation in same-generation ratios).
[NC-F12.3] This paper does NOT claim that the V↔F outer involution and the $\varepsilon \leftrightarrow -\varepsilon$ Z_2 are the same object. They are mathematically distinct (T_d outer automorphism vs. canonical J_Z register projection). What this paper claims is that BOTH trace to the same upstream polyhedral fact V(Tet) = F(Tet) = 4.
[NC-F12.4] This paper does NOT close the directional assignment OPEN of ZS-S9 §2 v1.0(Revised) (electron to V_XZ vs. V_ZY). The V↔F orientation in Theorem TDO-1 operates at the level of dimensionless ratio observables and does not require directional fixing of individual particle/antiparticle assignment, which remains an SSB-level OPEN problem.
[NC-F12.5] The 0.8% face-counting-vs-holonomy gap is NOT closed by this paper. It is registered as F-F12.6 OPEN with the $1/Q^2 \approx 0.83\%$ candidate explanation noted but not derived.
[NC-F12.6] This paper does NOT extend Layer Separation to instances outside the six enumerated in Table 6.1. The framework is presented as a HYPOTHESIS-strong meta-pattern with explicit instance verification; extension to additional Z-Spin observables is registered as F-ZTransform-1 OPEN (sub-gates 1a, 1b, 1c).
[NC-F12.7] The Cross-Coupling Theorem inheritance (ZS-M2 §5) used in §4.2 Step 1 is taken as PROVEN; this paper does not re-derive it or extend it.

§10. Conclusion

This paper has established two new theorems and one explicit negative result that together resolve the structural origin of the factor 2 in the 2e^A duality identity (Ω_Λ/Ω_m = m_d/m_u = 2.1668).

Theorem TDO-1 (Tetrahedral Dual Orientation Multiplicity, DERIVED-CONDITIONAL) proves that $\mu_{Tet} = 2$ is the multiplicity of oriented dual transport channels in the self-dual tetrahedron, forced by $V(Tet) = F(Tet) = 4$ (PROVEN, ZS-F9 Lemma 3.2) and unique to Tet among the five Platonic solids. The 4+4 self-referential face split of t-Tet (PROVEN, ZS-F9 §4 Observation 4.1) is the polyhedral signature of this two-fold orientation availability.

Theorem LSH-1 (Layer Separation Hypothesis, DERIVED-CONDITIONAL) provides a systematic factorization $R = \mu_{Tet} \times Hol_\varepsilon \times \mathscr{S}_{scale}$ for any dimensionless ratio observable mediated by the self-dual tetrahedron Z-sector, with $\mu_{Tet} = 2$ (Theorem TDO-1) and $Hol_\varepsilon = \exp(A)$ (ZS-F4 §6) as universal layers and $\mathscr{S}_{scale}$ as a scale-specific internal mechanism that cancels in same-class ratios.

The negative result F-Tet-2eA-1 REJECTED clarifies that V↔F outer involution and $\varepsilon \leftrightarrow -\varepsilon$ Z_2 are mathematically distinct objects, with V↔F entering $2e^A$ as a multiplicative prefactor outside the holonomy exponent (not as additive contribution inside, which would yield the falsified prediction $\exp(2A) \neq 2.1668$).

Six instances of the Scale-Invariant Z-Transform are enumerated from the corpus (Table 6.1), each verified to satisfy the LSH-1 factorization. The universal layered structure realizes "Unity Across Sixty Orders of Magnitude" (The Book §0.2) as a systematic mathematical structure: Layer 1 (polyhedral signature, $\mu_{Tet} = 2$ from $V(Tet) = F(Tet)$) and Layer 2 (ε holonomy, $\exp(A)$ from $\varepsilon \leftrightarrow -\varepsilon$ Z_2 doubling) are universal across all instances, while $\mathscr{S}_{scale}$ captures scale-specific physics.

ZS-S15 NC-S15.6 ("each factor 2 has its own derivation chain") is honored at a deeper level: the structural connection IS established at the polyhedral source ($\dim(Z) = 2$ from $V(Tet) = F(Tet)$), but each scale-specific manifestation retains its own derivation chain (SO(3)/SU(2) for Maxwell, $\Delta N_{eff} = 2A$ for cosmology, $\mu_{Tet} = 2$ in $2e^A$ ratios).

This paper introduces zero new free parameters; all inputs are LOCKED, PROVEN, or DERIVED in prior corpus papers (Table 2.1). Verification: 20/20 PASS across nine categories. Seven falsification gates F-F12.1 through F-F12.7 are pre-registered (Table 8.1). Three currently OPEN gates (F-F12.5, F-F12.6, F-F12.7-OPEN-direction) are explicitly tracked for future closure.

Acknowledgements & Code Availability

This work was developed with the assistance of AI tools (Anthropic Claude) for mathematical verification, character-theoretic computation, and manuscript drafting. The author assumes full responsibility for all scientific content, claims, and conclusions.

The verification suite is publicly available.
Verification script: zs_f12_verify_v1_0.py.
Dependencies: Python 3.10+, numpy, mpmath.
Execution: python3 zs_f12_verify_v1_0.py
Expected output: 20/20 PASS, exit code 0.

Appendix A. Verification Suite Results

All 20 tests pass at machine precision.

Table A.1. Verification suite results for zs_f12_verify_v1_0.py.

Cat.	Content	Tests	Pass/Fail
[A]	Locked Inputs (A, Q, (Z,X,Y), V(Tet)=F(Tet))	4	4 PASS
[B]	V/F Decomposition under T_d (A_1 ⊕ T_2)	3	3 PASS
[C]	μ_Tet = 2 Derivation (Theorem TDO-1)	2	2 PASS
[D]	μ_P = 1 for non-self-dual P (uniqueness)	4	4 PASS
[E]	LSH-1 Factorization Check (six instances)	3	3 PASS
[F]	Numerical 2e^A vs PDG/Planck Pulls	2	2 PASS
[G]	F-Tet-2eA-1 REJECTED Verification (exp(2A) ≠ 2e^A)	2	2 PASS
Total	All categories combined	20	20 PASS (100%)

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§11. Dated Update Annotations [v1.0(Revised), April 26, 2026]

This section records eight dated update annotations integrated as additive content per the ZS corpus no-deletion convention. The updates respond to a peer-review pass (April 26, 2026) that identified six core weaknesses in the v1.0 manuscript plus two consistency issues self-identified during the review. No prior numerical claim is modified; no new free parameter is introduced; no new physical prediction is added. Two theorem statuses are precision-refined and one application is honestly downgraded. External label v1.0(Revised) is adopted per the ZS-A8 v1.0 Revised precedent of accumulating in-place dated updates without version cascade. All §1–§10 content of v1.0 remains valid and unchanged; the present updates supplement v1.0 with sharpened formulations.

§11.0 Summary of Eight Dated Update Items

DU-1: Title subtitle precision ("Universal Factorization Theorem" → "Layer-Separation Principle").
DU-2: Verification Summary downgrade ("Two New Theorems" → "One DERIVED-CONDITIONAL theorem candidate + one HYPOTHESIS-strong layer principle").
DU-3: Abstract first paragraph honest reframing (cosmic $\Omega_\Lambda/\Omega_m = 83/38$ differs from holonomy $2e^A$ by 0.8% finite-Q discrepancy, registered as F-F12.6 OPEN).
DU-4: Epistemic Status Legend split (PROVEN and VERIFIED separated per The Book Table 0.3 and ZS-S15 corpus convention).
DU-5: Theorem TDO-1 proof reconstruction (Hom-space dimension 2 vs oriented duality groupoid multiplicity 2 distinguished; Orientation-counting convention added before Step 5; uniqueness via same-object endo-duality clarified in Step 6).
DU-6: Theorem LSH-1 status downgrade (Theorem → Principle; conditions (C4) explicit $\mathscr{S}$_scale construction and (C5) cancellation verification added).
DU-7: §5.2 Model-P claim correction (">5σ falsified" replaced with "1.7σ disfavored under PDG uncertainty"; numerical computation verified).
DU-8: §6 Six Z-Transform instances reorganized into three classes (Class I: exact $2e^A$ factorization, Class II: reciprocal Q/A ↔ A/Q, Class III: broader Z-Transform pattern). Pattern-mining critique addressed.

§11.1 Item DU-1: Title Subtitle Precision

v1.0 subtitle: "Polyhedral Origin of the Factor 2 in the $2e^A$ Identity, with a Universal Factorization Theorem for Scale-Invariant Z-Transform Observables."

v1.0(Revised) subtitle (operative for external citation): "Polyhedral Candidate for the Factor 2 in the $2e^A$ Identity, with a Layer-Separation Principle for Scale-Invariant Z-Transform Observables." The change "Origin → Candidate" honors the DERIVED-CONDITIONAL status of TDO-1 (the polyhedral source is established as candidate explanation, not as exclusive origin). The change "Universal Factorization Theorem → Layer-Separation Principle" honors the HYPOTHESIS-strong status of LSH-1 (see DU-6). The original v1.0 title remains valid as the internal label; the v1.0(Revised) subtitle is the form recommended for external citation and abstract reproduction.

§11.2 Item DU-2: Verification Summary Downgrade

v1.0 line: “Verification: 20/20 PASS | Zero Free Parameters | Two New Theorems (Theorem TDO-1, Theorem LSH-1).”

v1.0(Revised) operative line: “Verification: 20/20 arithmetic and structural checks PASS | Zero Free Parameters | One DERIVED-CONDITIONAL theorem candidate (Theorem TDO-1) and one HYPOTHESIS-strong layer principle (Principle LSH-1).” The 20 tests of Appendix A.1 remain unchanged; only the linguistic claim is honored to reflect the actual epistemic status of TDO-1 (DERIVED-CONDITIONAL, see DU-5) and LSH-1 (HYPOTHESIS-strong, see DU-6). This sharpens the reader-facing summary without modifying any verification result.

§11.3 Item DU-3: Abstract First-Paragraph Honest Reframing (Most Important Update)

The v1.0 Abstract opens with: “Two independent, corpus-PROVEN cosmological/quark identities share the same closed form: Ω_Λ/Ω_m = m_d/m_u = 2e^A = 2.1668...” This statement is in tension with §5.1 of the same paper, which honestly reports a 0.8% gap between the cosmic face-counting prediction (Ω_Λ/Ω_m = 83/38 = 2.1842) and the holonomy prediction (2e^A = 2.1668). The Abstract was written under the assumption that both ratios are exactly 2e^A, but the updated face-counting cosmic budget gives 83/38, not 2e^A.

v1.0(Revised) operative first paragraph: “Two corpus-level Z-Spin identities, one exact within the quark-sector Model-E chain (m_d/m_u = 2e^A = 2.1668, ZS-A5 §5.4 working hypothesis) and one approximate within the updated face-counting cosmic budget (Ω_Λ/Ω_m = 83/38 = 2.1842, ZS-A5 §3 DERIVED), converge on the same closed form 2e^A = 2.1668 up to a finite-Q discrepancy of order 1/Q^2 ≈ 0.83% on the cosmic side. This paper interprets the mismatch as an OPEN finite-Q correction (registered as F-F12.6), not as a closed theorem. Both ratios are extracted from physical scales separated by approximately forty orders of magnitude. The exp(A) factor is rigorously DERIVED in ZS-F4 §6 from the polyhedral Wilson-loop holonomy with $\varepsilon \leftrightarrow -\varepsilon$ Z_2 doubling. The factor 2 has been PROVEN in ZS-S15 §5 as the SO(3)/SU(2) double-cover ratio for the Maxwell field, but its structural origin in the polyhedral Z-sector mediator has been left OPEN.”

This update removes the false claim that both ratios are exactly 2e^A. The cosmic ratio is 83/38, the holonomy prediction is 2e^A, and the 0.8% gap is honestly registered as OPEN. F-F12.6 records the candidate finite-Q correction ($1/Q^2 \approx 0.83\%$, 3% accuracy match), but the explicit derivation remains future work.

§11.4 Item DU-4: Epistemic Status Legend Split (PROVEN / VERIFIED Separation)

The v1.0 Epistemic Status Legend defines PROVEN as “Mathematical theorem with complete proof, or numerical verification at machine precision (≤ 10^(-10) residual).” This combines two distinct epistemic categories that the broader corpus separates. The Book Table 0.3 (PROVEN), ZS-S15 v1.0, ZS-M2 v1.0, and ZS-M8 v1.0 all distinguish PROVEN (mathematical theorem) from VERIFIED (numerical confirmation).

v1.0(Revised) refined definitions (operative for all subsequent ZS-F12 dated updates): PROVEN: Mathematical theorem with complete proof. Cannot be falsified by experiment, only by logical error. VERIFIED: Numerical confirmation of a PROVEN or DERIVED result to machine or 50-digit precision. Strengthens confidence but is not a proof. The Locked Inputs Table 2.1 entries previously labeled PROVEN with numerical verification (such as 2e^A duality “VERIFIED”) retain their classifications.

The Table 8.1 falsification gates that read “PROVEN PASS” should be read as “VERIFIED PASS” where the verification is numerical (F-F12.1, F-F12.2, F-F12.3, F-F12.4); only F-F12.1 carries genuine PROVEN status by inheriting the PROVEN status of ZS-F9 §3.2 Lemma 3.2 (Mathematical theorem).

§11.5 Item DU-5: Theorem TDO-1 Proof Reconstruction

The v1.0 §3.2 proof of Theorem TDO-1 conflates two distinct “2”s. Step 2 invokes Schur’s lemma to compute $\dim \mathrm{Hom}_{T_d}(A_1 \oplus T_2, A_1 \oplus T_2) = 1 + 1 = 2$ (the Hom-space dimension, arising from two irreducible summands). Step 5 then claims $\mu_{Tet} = 2$ from V→F and F→V orientations (the dual-orientation count). These two “2”s are formally distinct mathematical objects: the Hom-space dimension counts intertwiner algebra basis elements, while the dual-orientation multiplicity counts directed morphisms in the duality groupoid. The v1.0 proof does not establish that the two “2”s coincide.

Reconstructed §3.2 (operative): Replace Step 2 and Step 5 with the following Orientation-Counting Convention.

Orientation-Counting Convention (operative for v1.0(Revised)). The multiplicity μ_{Tet} is NOT defined as the dimension of the T_d-equivariant endomorphism algebra $\mathrm{Hom}_{T_d}(A_1 \oplus T_2, A_1 \oplus T_2)$. Instead, μ_{Tet} is defined in the oriented duality groupoid $\mathscr{D}_{or}$ of the self-dual cell complex Tet, where $\mathscr{D}_{or}$ has two generating morphisms $\mathscr{D}_{VF}: C_0(Tet) \to C_2(Tet)$ and $\mathscr{D}_{FV}: C_2(Tet) \to C_0(Tet)$ satisfying $\mathscr{D}_{FV} \circ \mathscr{D}_{VF} = id_{C_0}$ and $\mathscr{D}_{VF} \circ \mathscr{D}_{FV} = id_{C_2}$. Since Tet ≃ Tet*, both morphisms close inside the same object (same-object endo-duality). The multiplicity is counted as $\mu_{Tet} = |\mathrm{Hom}_{\mathscr{D}_{or}}(C_0, C_2)| + |\mathrm{Hom}_{\mathscr{D}_{or}}(C_2, C_0)| = 1 + 1 = 2$. Schur’s lemma is then used in an auxiliary capacity only: to establish that the two oriented morphisms $\mathscr{D}_{VF}$ and $\mathscr{D}_{FV}$ are T_d-equivariantly weighted equally (each generates a single intertwiner line in the relevant Schur channel), which guarantees that the multiplicity-2 contribution is balanced rather than skewed.

Reconstructed Step 6 (uniqueness via same-object endo-duality, operative for v1.0(Revised)). For a non-self-dual Platonic solid P (cube, octahedron, dodecahedron, icosahedron), the duality functor $\mathscr{D}: C_{\cdot}(P) \to C_{\cdot}(P^*)$ maps to a distinct object $P^* \neq P$. Although P and P* together form a dual pair admitting two oriented directions ($P \to P^*$ and $P^* \to P$), these directions do not close inside the same polyhedral object; they require simultaneous tracking of two distinct cell complexes. For dimensionless ratio observables that probe the same-object Z-mediator (where the source and target are the same polyhedral object Tet), only the Tet endo-duality provides the multiplicity-2 structure within a single object. Non-self-dual solids contribute multiplicity 1 in the same-object sense, because the second direction lives in a different polyhedral object. Hence $\mu_P = 1$ (within-object) for non-self-dual P, while $\mu_{Tet} = 2$ is unique to the regular tetrahedron, the unique self-dual Platonic solid.

Status preservation. Theorem TDO-1 retains its DERIVED-CONDITIONAL status. The conditions are now refined: (a) standard interpretation of polyhedral duality as a functor of cell complexes; (b) identification of V↔F orientations with morphisms in the oriented duality groupoid $\mathscr{D}_{or}$, not with intertwiners in the endomorphism algebra; (c) restriction to dimensionless ratio observables that probe same-object endo-duality.

§11.6 Item DU-6: Theorem LSH-1 → Principle LSH-1 (Status Downgrade)

The v1.0 §4 calls LSH-1 a “Theorem.” Inspection of the proof sketch reveals that it is, more accurately, a factorization ansatz: Step 5 states that “the total observable R must contain both layers multiplicatively,” without proving this multiplicative form. The $\mathscr{S}$_scale cancellation in same-class ratios is asserted but not

derived for each instance. F-F12.5 OPEN in the v1.0 falsification table acknowledges this. Per ZS corpus convention (ZS-S13 §6.16, ZS-M11 §H16, ZS-A6 NC-A6.2), DERIVED-CONDITIONAL strong is the strongest status applicable to a result that requires per-instance cancellation verification.

v1.0(Revised) operative status: LSH-1 is reclassified from "Theorem" to "Principle" with HYPOTHESIS-strong status. The qualifier "HYPOTHESIS-strong" reflects: multiple independent lines of evidence (six instances, Cross-Coupling Theorem inheritance, Block Fiedler reciprocal duality), but a complete derivation chain that does not depend on per-instance $\mathscr{S}$_scale verification is not yet established. Two new conditions are added to the statement of LSH-1, operative for v1.0(Revised) external citation.

Principle LSH-1 (operative for v1.0(Revised), HYPOTHESIS-strong). Let R denote a dimensionless ratio observable satisfying conditions (C1)–(C3) of v1.0 §4.1, plus the following two refined conditions: (C4) $\mathscr{S}$_scale is explicitly constructed for the specific observable R, identifying the scale-specific internal mechanism (e.g., D_5 rotation for quark Yukawa, face counting for cosmic budget); (C5) the same-class cancellation $\mathscr{S}$_scale^(X) / $\mathscr{S}$_scale^(Y) ≈ 1 is verified independently for the ratio under consideration. Then R admits the universal factorization R = μ_Tet × Hol_ε × residual where residual = $\mathscr{S}$_scale^(X) / $\mathscr{S}$_scale^(Y) is the residual scale-specific mechanism. For ratios where (C4)–(C5) hold with residual = 1 exactly, R = μ_Tet × Hol_ε = 2e^A.

Per-instance verification status: m_d/m_u (Class I) verified at residual ≈ 1 within Schur conservation Σσ_i² = 1/5 (ZS-M10 §3.4 PROVEN, same-generation cancellation argument of v1.0 §5.2 retained). Ω_Λ/Ω_m (Class I, approximate) verified at residual = 1 + ε with ε ≈ 1/Q² ≈ 0.83% (F-F12.6 OPEN, candidate finite-Q correction). Class II and Class III instances of the reorganized §6 (DU-8) are NOT claimed to satisfy the strict (C4)–(C5) form; they share Layer 1 and Layer 2 universal factors but not the closed form 2e^A.

§11.7 Item DU-7: §5.2 Model-P Numerical Correction

v1.0 §5.2 states: "Model-P (m_d/m_u = 2e^(√3·A) = 2.298) is FALSIFIED at >5σ." Direct arithmetic check using the PDG uncertainty ±0.08 quoted in the same paragraph: |2.298 − 2.16|/0.08 = 0.138/0.08 = 1.725. Hence Model-P is 1.7σ above the PDG central value, not >5σ. The ">5σ" figure was inherited from ZS-A5 v1.0 §5.4 without independent verification.

v1.0(Revised) operative §5.2 statement (replacing the ">5σ falsified" claim): "Model-P (m_d/m_u = 2e^(√3·A) = 2.298) is disfavored by approximately 1.7σ relative to PDG 2.16 ± 0.08. Model-E (m_d/m_u = 2e^A = 2.1668) is preferred at the 0.31% pull level. Under the larger uncertainties typical of lattice-QCD m_d/m_u extractions (FLAG 2024 typically quotes ±0.05–0.10 depending on scheme), the Model-P versus Model-E discrimination is at the 1.4–2.8σ level. A definitive lattice average with σ ≤ 0.03 would discriminate at >3σ; this is registered as a refined falsification gate F-F12.3-refined."

Cross-paper note: ZS-A5 v1.0 §5.4 also contains the ">5σ" phrasing. A coordinating dated update on ZS-A5 §5.4 is recommended; this is registered as cross-paper consistency item to be addressed in the next ZS-A5 dated update batch.

§11.8 Item DU-8: §6 Six Instances Three-Class Reorganization

v1.0 §6 Table 6.1 lists six Z-Transform instances with the umbrella claim that "each is verified to satisfy LSH-1 factorization." However, the six instances do not all share the same closed form: instances 1a, 1b have form 2e^A; instances 2a, 2b have forms Q/A and A/Q (reciprocal pair); instances 3a, 3b, 4a, 4b,

5a–c, 6a–c have other forms ($y_t \cdot v \cdot (A/Q)$, $|z^*|^2$, various $\exp(n\pi/A)$). Treating all six as "LSH-1 verified" is over-extension and risks pattern-mining critique.

v1.0(Revised) operative reorganization: The six instances of v1.0 Table 6.1 are partitioned into three classes with explicitly distinct claims.

Class I (exact 2e^A factorization, central LSH-1 claim). $m_d/m_u = 2e^A$ (exact within Model-E working hypothesis), $\Omega_\Lambda/\Omega_m \approx 2e^A$ (approximate, 0.8% finite-Q gap registered as F-F12.6 OPEN). Both observables admit explicit (C4)–(C5) verification of LSH-1 with residual ≈ 1. Status: DERIVED-CONDITIONAL (m_d/m_u within Model-E) and DERIVED-CONDITIONAL (Ω_Λ/Ω_m up to F-F12.6).

Class II (reciprocal impedance pair, related but not 2e^A). $1/\alpha_{EM} \approx Q/A$ (X-face, ZS-A1 T1-2), $\varepsilon_{solar} \approx A/Q$ (Y-face, ZS-S2 T1-3). These are reciprocal forms of the Block Fiedler eigenvalue $\lambda_2 = 2A/Q$ (PROVEN, ZS-T1 §9.3), not instances of the closed form 2e^A. The factor 2 in $\lambda_2 = 2A/Q$ does share the polyhedral origin dim(Z) = 2 (same upstream as μ_{Tet}), but the holonomy factor exp(A) is absent from these instances. Status: PROVEN within ZS-T1 framework, included in v1.0(Revised) §6 only to illustrate the broader Z-Transform pattern of Layer 1 (factor 2 from polyhedral signature) inheritance.

Class III (broader Z-Transform pattern, no exact 2e^A claim). Hubble three-level holonomy ($H_0/\sqrt{(1+A)}$, H_0, $H_0 \cdot \exp(A)$), timescale hierarchy $\tau_n = t_P \cdot \exp(n\pi/A)$, $\eta_{topo} \approx \Omega_m$ near-equality, m_τ register/spectral faces (ZS-S8 H1/H2). These instances exhibit Layer 2 (holonomy factor exp(A) or $\exp(n\pi/A)$) but not the universal Layer 1 × Layer 2 = 2e^A combination. They are included in v1.0(Revised) §6 only as evidence that the broader layered structure (some combination of polyhedral signature and ε holonomy) recurs across the corpus. They are NOT claimed to satisfy the strict (C4)–(C5) form of Principle LSH-1. Status: OBSERVATION (pattern recognition, no strict factorization claim).

Operative reading. The central LSH-1 claim is restricted to Class I instances. Class II provides upstream evidence for the Layer 1 factor 2 (sharing polyhedral origin), and Class III provides broader corpus evidence that the layered structure recurs without exact 2e^A reproduction. The pattern-mining critique is addressed by separating "LSH-1 is verified" (Class I) from "the Z-Transform pattern is observed" (Class II + III).

§11.9 Verification Suite Reconciliation

The v1.0 verification suite zs_f12_verify_v1_0.py (20/20 PASS, Appendix A) remains valid; no test result is invalidated by the v1.0(Revised) updates. Categories [A]–[G] of Table A.1 retain their PASS status. The eight dated updates of §11.1–§11.8 are linguistic/structural refinements at the manuscript level (subtitle wording, status labels, explicit cancellation conditions, classification clarity, numerical correction, Z-Transform instance grouping); they do not modify any of the 20 numerical or structural identities tested in the verification suite. A new audit script zs_f12_revised_audit_v1_0.py is registered (PLANNED) to add 8 dated-update consistency tests (one per DU item), targeting a total combined verification of 20 + 8 = 28 PASS at the v1.0(Revised) level.

§11.10 Net Effect of v1.0(Revised) Updates

After application of the eight dated updates (DU-1 through DU-8), the operative external citation form of ZS-F12 is: "Tetrahedral Dual Orientation Multiplicity and the Layer Separation Hypothesis (Polyhedral Candidate for the Factor 2 in the 2e^A Identity, with a Layer-Separation Principle for Scale-Invariant

Z-Transform Observables), Z-Spin Cosmology v1.0(Revised) (April 2026).” The central scientific content is preserved unchanged: Theorem TDO-1 establishes μ_Tet = 2 from tetrahedral V↔F self-duality (DERIVED-CONDITIONAL); Principle LSH-1 establishes the Layer 1 × Layer 2 × $\mathscr{S}$_scale factorization for Class I observables (HYPOTHESIS-strong); F-Tet-2eA-1 REJECTED with replacement gate F-Tet-2eA-2 active. The negative result and the new theorem candidate are both honored at honest epistemic levels, the 0.8% face-counting-vs-holonomy gap is registered as F-F12.6 OPEN throughout the manuscript, and the pattern-mining critique is addressed by Class I/II/III separation. Zero new free parameters introduced; all v1.0 numerical claims and verification PASS counts preserved.

Methodological lesson recorded for the corpus: the v1.0 manuscript passed the standard 9-step Z-Spin verification protocol but did not survive an explicit external-critique simulation. The peer-review pass identified six core weaknesses (Hom-space vs orientation conflation, non-self-dual μ=1 weakness, LSH-1 status overclaim, Abstract–§5.1 internal conflict, Model-P numerical error, Class I/II/III conflation) that the internal protocol did not flag. A future addition to the standard 9-step protocol (proposed Step 5.5: explicit external-critique simulation) is registered as ZS corpus-wide methodology improvement.

Version History

v1.0 (April 2026): Initial public release. Contents: Theorem TDO-1 (Tetrahedral Dual Orientation Multiplicity, DERIVED-CONDITIONAL), Theorem LSH-1 (Layer Separation Hypothesis, DERIVED-CONDITIONAL), F-Tet-2eA-1 REJECTED with new gate F-Tet-2eA-2 registered. Six Z-Transform instances enumerated (Table 6.1) with universal layered structure Layer 1 × Layer 2 × $\mathscr{S}$_scale. Seven falsification gates F-F12.1 through F-F12.7 pre-registered. Seven non-claims NC-F12.1 through NC-F12.7 documented. Verification suite zs_f12_verify_v1_0.py with 20/20 PASS at machine precision. Anti-numerology Monte Carlo planned (zs_f12_mc_v1_0.py). Zero new free parameters; all inputs LOCKED, PROVEN, or DERIVED in prior corpus papers. Word count strictly increased relative to all referenced upstream papers per the v1.0 freeze convention. (Consolidated from internal Z-Spin Collaboration research notes through April 2026.)

v1.0(Revised) (April 26, 2026): Eight dated update annotations integrated as additive content per the no-deletion rule (§11.1–§11.8). Status precision-refinements: Theorem TDO-1 retains DERIVED-CONDITIONAL with reconstructed proof distinguishing Hom-space dimension from oriented duality groupoid multiplicity (DU-5); Theorem LSH-1 reclassified Theorem → Principle with HYPOTHESIS-strong status and added conditions (C4)–(C5) for explicit $\mathscr{S}$_scale construction and same-class cancellation verification (DU-6). Manuscript-level corrections: subtitle “Universal Factorization Theorem” → “Layer-Separation Principle” (DU-1); Verification Summary “Two New Theorems” → “One DERIVED-CONDITIONAL theorem candidate + one HYPOTHESIS-strong layer principle” (DU-2); Abstract first paragraph rewritten to honor the 0.8% face-counting-vs-holonomy gap as F-F12.6 OPEN (DU-3); Epistemic Status Legend split PROVEN/VERIFIED per The Book Table 0.3 convention (DU-4); §5.2 Model-P claim corrected from “>5σ falsified” to “1.7σ disfavored under PDG uncertainty” (DU-7); §6 six Z-Transform instances reorganized into Class I (exact 2e^A), Class II (reciprocal Q/A), Class III (broader pattern) per pattern-mining critique (DU-8). New verification audit zs_f12_revised_audit_v1_0.py registered (PLANNED, 8 dated-update consistency tests). Methodological lesson recorded: peer-review pass identified six core weaknesses not flagged by the standard 9-step ZS verification protocol; future protocol addition (proposed Step 5.5: external-critique simulation) registered as corpus-wide methodology improvement. No prior numerical claim modified; no new free parameter;

no new physical prediction. v1.0 verification 20/20 PASS preserved. External label v1.0(Revised) adopted per ZS-A8 v1.0 Revised precedent of accumulating in-place dated updates without version cascade. Word count strictly increased (+~3500 words from §11 dated update section).